RABIMBA KARANJAI

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PARTICULARS

EDUCATION

University Of Houston — Houston, TX
Ph. D. in Computer Science — *2020 - 2025 (Defended July,25)*
Industry Work Experience: 2019-2020
Rice University — Houston, TX
M. S. in Computer Science — *2015 - 2018*

RESEARCH INTERESTS

My research focuses on trustworthy and agentic artificial intelligence, large language model reasoning, code generation, and AI-assisted software security. I develop methods for improving the reliability, safety, and verifiability of LLM-based systems, with particular emphasis on smart contracts, decentralized systems, formal verification, and automated program repair. I also investigate applications of these methods in financial systems, cybersecurity, and biomedical research.

DISSERTATION

Title: "Teaching AI's to reason and Code, Confidentially" — *Best Dissertation Award*
Advisor: Prof. Weidong (Larry) Shi

ACADEMIC HONORS

- ACM SIGSOFT Distinguished Paper Award, 2025
- Dan E. Wells Outstanding Dissertation Award - University Of Houston, Fall 2025
- Outstanding PhD Student - University Of Houston, 2022-2023, 2023-2024
- Graduate Tuition Fellowship, University Of Houston, 2021-2025.
- Google Cloud Research Innovator, 2023.
- Cullen Travel Fellowship, 2023.
- Google Research CS research mentorship program, 2023.
- HPE DSI Artificial Intelligence Summer Student Showcase - 3rd Place, 2023-2024
- Distinguished Paper Candidate, IEEE ICBC 2022.
- Mozilla Research Fellowship, 2018.
- Graduate Tuition Fellowship, Rice University, 2015-2018.

WORK EXPERIENCE

- **Staff Agentic AI Researcher, PayPal Research, San Jose, CA,** Sep 2025 – Present. Lead the technical strategy and deployment of production-scale agentic AI systems, utilizing massive-scale multimodal data to optimize user engagement and financial applications. Spearheaded the end-to-end development of an in-house voice-to-voice model in partnership with NVIDIA (featured in an upcoming NVIDIA publication) and built a frontier multimodal model in collaboration with Google(lead author in both). Designed and implemented an agentic development loop using novel preference optimization frameworks (CM-DPO), improving model planning and ranking under real-world constraints. Built and refined automated training, evaluation, and feedback pipelines for frontier LLMs (up to 120B parameters), continuously learning from offline data and production behavior. Mentored senior technical talent and drove multi-quarter roadmaps, setting modeling best practices and leading cross-functional AI initiatives. Designed a multi-agent adversarial evaluation arena to benchmark system robustness, ensuring models meet strict reliability and safety bars. Filed 3 U.S. patents in emerging AI technologies at PayPal.
- **Senior Software Engineer (Machine Learning), Clearedin,** Dec 2020 - Aug 2021. Designed and implemented advanced ML models and threat detection engines for high-throughput email scanning pipelines using Python and C++. Spearheaded the development of machine learning pipelines for large-scale classification and ranking of malicious emails, utilizing deep learning and gradient boosted trees to significantly enhance accuracy. Partnered with product and infra teams to translate threat intelligence into reliable, scalable production models, evaluating online behavior and continuous experimentation results.
- **Machine Learning Lead, Fireflies.ai.**, Aug. 2020 - Dec 2020. Led the technical design and deployment of large-scale ML models for conversational intelligence, serving over 10 million users globally. Built the first Machine Learning team from the ground up, mentoring MLEs and establishing best practices for model training, offline evaluation, and reliable production deployment. Leveraged modern data and experimentation platforms to iteratively improve speech and NLP model relevance, translating business objectives into a multi-quarter ML roadmap.
- **Programmer Analyst, Cognizant Technology Solutions,**, Dec. 2010 - Aug. 2013. Contributed to the Innovation team by spearheading the development of automated mobile hardware and software testing solutions. Leveraged OpenCV and vision-based technologies to design advanced testing frameworks, enabling efficient and reliable validation of mobile devices.

RESEARCH EXPERIENCE

- **Research Assistant, University Of Houston**, Sep 2021 - August 2026.
  AI & Software Engineering: I explored the potential of artificial intelligence (AI) in Code Generation, Low Resource Code translation and Reasoning. I also explored it's use in healthcare by developing a system for AI-driven analysis of histopathology images to enhance vascular disease diagnosis. Additionally, I investigated the challenges of mitigating hallucinations in AI-driven medical diagnosis, contributing to the development of more reliable AI systems in healthcare. I also examined the irrationality in large language models (LLMs) and identified open research questions, leading to a better understanding of the limitations and potential improvements of LLMs. This research demonstrates my efforts in advancing AI and LLM technologies for practical applications.

  Decentralization: I examined the possibility of creating decentralized infrastructure as code (IaC) using blockchain technology, which led to the development of DlaC, a novel system that ensures secure and transparent management of infrastructure. I designed and implemented a decentralized function-as-a-service (FaaS) framework over multiple clouds, enhancing the efficiency and scalability of decentralized applications (dApps) and Web3 services. I also explored the use of trusted execution environments (TEEs) for critical infrastructure protection, enabling secure and private computation in blockchain applications. This work contributes to the advancement of decentralized technologies and their application in various domains.
- **Research Assistant, Rice University**, Jan 2015 - Aug 2018.
  High Performance Computing: Worked on java script parallelization on Mozilla Firefox and other web browser engines so that it can run on weareable VR headsets.
- **Research Associate, Mozilla Research,** Sep. 2017 - Jan. 2018. Conducted in-depth research to optimize parallelism and performance in Web Virtual Reality and WebGL, focusing on enhancing the efficiency of three.js, which became my masters thesis. Developed techniques to offload resource-intensive computations to the GPU using WebAssembly, significantly improving performance and reducing latency in VR applications. Parallelized object loading and texture rendering in three.js through non-blocking GPU processes, enabling faster and more efficient rendering.
- **Summer Research Intern, IBM Almaden Research Center,** May. 2017 - Aug. 2017. Developed and got awarded a patented multi-factor authentication system (US Patent US11025643B2) integrating biometric features and blockchain technology. Designed identity specifications for Hyperledger Fabric to enable secure and reliable digital identity management. Innovated Android-based multi-party fingerprint authentication and iris recognition using cancellable biometrics, enhancing both security and privacy. Established methods for extracting fingerprint templates and features on commodity mobile devices, improving accessibility and convenience in biometric authentication solutions.

- **Summer Research Intern, IBM Almaden Research Center,** May. 2015 - Aug. 2015. Developed a tamper-evident mobile application solution capable of provably verifying application integrity, even on rooted or jailbroken devices. Designed and implemented a tamper-evident data structure and logging system leveraging ARM TrustZone and blockchain technology to securely store and log data, ensuring resilience against tampering and unauthorized modifications.
- **Researcher, IBM T J Watson Research Lab,** May. 2014 - Nov. 2014. Embedded cognitive computing and NLP into real-world Watson solutions by developing advanced systems for data extraction, processing, and analysis. Designed a document parsing system utilizing OCR and DeepQA, training models on extensive datasets and optimizing performance for accurate data extraction. Engineered a scalable solution for storing and processing massive datasets, including the development of user-friendly front-end and visualization libraries. Created and trained customized Natural Language Processing models and evidence-scoring systems, optimized for self-learning and tailored to specific use cases.

TEACHING EXPERIENCE

- **Instructional Assistant.** COSC 6309: Introduction to Automata and Computability, Fall 2025, University Of Houston.
- **Instructional Assistant.** COSC 4351: Fundamentals of Software Engineering, Spring 2022, Summer 2022 and Summer 2023, Summer 2024, University Of Houston.
- **Instructional Assistant.** COSC 2306: Data Programming, Fall 2023, University Of Houston.
- **Instructional Assistant.** COSC 6376: Cloud Computing, Fall 2022, Spring 2023, University Of Houston.
- **Teaching Assistant.** COMP 541: Intro to Computer Security, Spring 2016, Rice University.
- **Teaching Assistant.** COMP 321: Introduction to Computer Systems, Fall 2016, Rice University.
- **Teaching Assistant.** COMP 424: Mobile and Embedded Systems, Spring 2017, Rice University.

PUBLICATIONS

PAPERS

1. **Rabimba Karanjai**, Yang Lu, Hemanth Hegadehalli Madhavarao, Lei Xu, Weidong Shi, "***Context Contamination in LLM Analysis of Network Security Logs: Poison with Passive Prompt Injection and Mitigation Evaluation***", *USENIX Security, 2026.*
2. **Rabimba Karanjai**, Yang Lu, Nour Diallo, Wujie Xiong, Lei Xu, Weidong (Larry) Shi, "***When Agents Act on Web3: An Attack-Surface Survey of MCP, Skills, and Tool Calling***," *To appear, BRAINS 2026.*
3. **Rabimba Karanjai**, Yang Lu, Richard Williamson, Hemanth Hegadehalli Madhavarao, Prakhar Mehrotra, Lei Xu, Weidong (Larry) Shi, "***PACE: Policy-Attested Contract Execution for Safe AI Agents in Decentralized Finance***," *To appear, BRAINS 2026.*
4. **Rabimba Karanjai**, Hemanth Hegadehalli Madhavarao, Lei Xu, Weidong Shi, "***CausalGraphX: A Counterfactual Graph Neural Network Framework for Explainable Systemic Risk Assessment***", *AI4Finance, AAAI 2026.*
5. **Rabimba Karanjai**, Hemanth Madhavarao, Lei Xu, Weidong Shi, "***QuCoWE: Quantum Contrastive Word Embeddings with Variational Circuits for Near-Term Quantum Devices***", *QC+AI, AAAI, 2026.*
6. **Rabimba Karanjai**, Lei Xu, Weidong Shi, " ***Securing the Multi-Chain Ecosystem: A Unified, Agent-Based Framework for Vulnerability Repair in Solidity and Move***", *to be presented at ACM AIWare 2025.* ***Winner of ACM SIGSOFT Distinguished Paper award.***
7. **Rabimba Karanjai,** "***Teaching AIs to Reason and Code, Confidentially***", PhD Thesis. — ***Best Dissertation Award - University Of Houston***
8. **Rabimba Karanjai**, Lei Xu, Weidong Shi, "***HPCAgentTester: A Multi-Agent LLM Approach for Enhanced HPC Unit Test Generation***", *to be presented at ACM AIWare 2025.*
9. **Rabimba Karanjai**, Yang Lu, Ranjith Chodavarapu, Lei Xu, Weidong Shi "***Evaluating the Quality of Randomness and Entropy in Tasks Supported by Large Language Models***,", *KDD - Agentic & GenAI Evaluation*, 2025.
10. **Rabimba Karanjai**, Sam Blackshear, Lei Xu, Weidong Shi "***Collaboration is all you need: LLM Assisted Safe Code Translation***,", *FSE*, 2025.
11. **Rabimba Karanjai**, Boris Shor, Amanda Austin, Ryan Kennedy, Yang Lu, Lei Xu, Weidong Shi, "***Synthesizing Public Opinions with LLMs: Role Creation, Impacts, and the Future to eDemorcacy***", *ICEDEG 2025.*
12. **Rabimba Karanjai**, Yang Lu, Dana Alsagheer, Keshav Kasichainula, Lei Xu, Weidong Shi, Stephen Huang, "***LogBabylon: A Unified Framework for Cross-Log File Integration and Analysis***,", *ACM SAC*, 2025.
13. Suravi Majumder, Koushik Sen, **Rabimba Karanjai**, "***AI-Based Target for Personalized Interventions of Atherosclerosis from Gut Microbiota Signature***,", *SynBio*, 2025.

14. **Rabimba Karanjai**, Lei Xu and Weidong Shi, "***Smart Contract Code Translation based on Concepts***,". *ACM Foundations of Software Engineering (FSE)*, 2024.
15. **Rabimba Karanjai**,Sangwon Shin, Wujie Xiong, Xinxin Fan, Lin Chen, Tianwei Zhang, Taeweon Suh,Weidong Shi, Veronika Kuchta, Francesco Sica and Lei Xu, "***TPU as Cryptographic Accelerator***", *MICRO*, 2024.
16. Ranjith Chodavarapu, **Rabimba Karanjai**, Xinxin Fan, Larry Shi and Lei Xu, "***Adding All Flavors: A Hybrid Random Number Generator for dApps and Web3***,", *International Symposium on Stabilization, Safety, and Security of Distributed Systems (SSS)*, 2024.
17. **Rabimba Karanjai** and Weidong Shi, "***Trusted LLM Inference on the Edge with Smart Contracts***", *IEEE ICBC*, 2024.
18. **Rabimba Karanjai**,Lei Xu,Lin Chen,Nour Diallo,Weidong Shi, "***Decentralized FaaS over Multi-Clouds***", *ACM SAC*, 2024.
19. Nour Diallo, Dana Alsagheer, Lei Xu, Yang Lu, **Rabimba Karanjai**, Weidong Shi and Mohammad Kamal, "***All We Need is Voter Feedback***",*IEEE ICEDEG*, 2024.
20. **Rabimba Karanjai**,Keshav Kasichainula,Lei Xu,Nour Diallo,Lin Chen,Weidong Shi, "***DIaC: Re-imagining Decentralized Infrastructure As Code using Blockchain***", *IEEE TNSM*, 2023.
21. **Rabimba Karanjai**, E Li, L Xu, W Shi, "***Who is Smarter? An Empirical Study of AI-based Smart Contract Creation***", *BRAINS*, 2023.
22. **Rabimba Karanjai**, Rowan Collier, Zhimin Gao, Lin Chen, Xinxin Fan, Taeweon Suh, Weidong Shi and Lei Xu, "***Supporting Heterogeneous TEE for Critical Infrastructure Protection***", *ACM AsiaCCS*, 2023.
23. **Rabimba Karanjai**, Keshav Kasichainula,Nour Diallo,Mudabbir Kaleem,Lei Xu,Lin Chen,Weidong Shi, "***Decentralized Application Infrastructures as Smart Contract Codes* — Distinguished Paper Award**",*IEEE ICBC*, 2022.
24. **Rabimba Karanjai**, Lei Xu,Zhimin Gao,Lin Chen,Mudabbir Kaleem,Weidong Shi, "***Privacy preserving event based transaction system in a decentralized environment***",*ACM Middleware*, 2021.
25. **Rabimba Karanjai**, Lei Xu, Zhimin Gao, Lin Chen, Mudabbir Kaleem, Weidong Shi, "***On Conditional Cryptocurrency With Privacy***",*IEEE ICBC*, 2021.
26. **Rabimba Karanjai**, L Xu, L Chen, F Zhang, Z Gao, W Shi, "***Lessons Learned from Blockchain Applications of Trusted Execution Environments and Implications***",*ACM HASP*, 2021.
27. Mudabbir Kaleem,Keshav Kasichainula,**Rabimba Karanjai**,Lei Xu,Zhimin Gao,Lin Chen,Weidong Shi, "***An event driven framework for smart contract execution***",*ACM DEBS*, 2021.
28. **Rabimba Karanjai**, "***Optimizing Web Virtual Reality***",*Masters Thesis*, Rice University, 2018.

SHORT PAPERS

29. **Rabimba Karanjai**, Yang Lu, Lei Xu and Weidong (Larry) Shi, "***Empowering Smart Contracts with Real-time On-Chain AI Inferences***", *IEEE International Conference on Blockchain and Cryptocurrency (ICBC) 2026.*
30. **Rabimba Karanjai**, Yang Lu, Lei Xu and Weidong (Larry) Shi, "***Unlocking On-Chain Intelligence: A Practical Framework for GenAI-Powered Smart Contracts***", *BRAINS 2025.*
31. Xinyu Hou, Yang Lu, **Rabimba Karanjai**, Lei Xu, Weidong Shi, "***Ransomware 3.0: Enhancing Risk Management and Mitigation Options with Proof-of-Decryptability and Smart Contracts***", *IEEE ICBC 2025.*
32. **Rabimba Karanjai**, Lei Xu ,Nour Diallo, Lin Chen and Weidong Shi, "***DeFaaS: Decentralized Function-as-a-Service for Emerging dApps and Web***", *IEEE ICBC*, 2023.
33. **Rabimba Karanjai**, Zhimin Gao, Lin Chen, Xinxin Fan, Taeweon Suh, Weidong Shi and Lei Xu, "***DHTee: Decentralized Infrastructure for Heterogeneous TEE***", *IEEE ICBC*, 2023.
34. Dana R Alsagheer, Nour Diallo, Lei Xu, **Rabimba Karanjai** and Weidong Shi, "***Decentralized Machine Learning Governance: Overview, Opportunities, and Research Challenges***", *IEEE ICBC*, 2023.

WORKSHOP PAPERS

35. **Rabimba Karanjai**, Suravi Majumder, "***Mitigating Hallucinations in AI-Driven Medical Diagnosis***", *3rd Annual AI in Health Conference*, Rice University, Ken Kennedy, 2024.
36. **Rabimba Karanjai**, Suravi Majumder, "***Enhancing Vascular Disease Diagnosis through AI-Driven Analysis of Histopathology Images***", *3rd Annual AI in Health Conference*, Rice University, Ken Kennaey, 2024.
37. Dana R Alsagheer,**Rabimba Karanjai**, Weidong Shi,Nour Diallo, Yang Lu, Suha Beydoun, Qiaoning Zhang, "***Evaluating Irrationality in Large Language Models and Open Research Questions***", *ACM CHI*, HEAL, 2024.

PAPERS UNDER REVIEW

38. Dana R Alsagheer, Yang Lu, **Rabimba Karanjai**, Lei Xu, Weidong Shi, "***Unanimous but Wrong: Conformity in Multi-Agent LLM Audit Deliberation***", *AAAI 2027.*
39. Yang Lu, **Rabimba Karanjai**, Dana R Alsagheer, Lei Xu, Weidong Shi, "***Motivated Chain-of-Thought and Reduced Scrutiny in LLM Judges***", *AAAI 2027.*
40. Xinyu Hou, Yang Lu, **Rabimba Karanjai**, Pei-Chi Pan, Sen Lin, Lei Xu, Weidong Shi, "***Designing the Outside Option: Anchors, Mixed Equilibria, and Welfare in LLM Routing Auctions***", *AAAI 2027.*
41. Jenna L. Ballard, **Rabimba Karanjai**, Yixi Li, Yun-Shiuan Chuang, Shivani Shekhar, Avinash Thangali, Yiming Bian, Yirou Ge, Uma Kona, Linsey Pang, Prakhar Mehrotra, "***TransactQA: Benchmarking and Modeling Conversational Question Answering over Irregular, Heterogeneous Transaction Sequences***", *AAAI 2027.*
42. **Rabimba Karanjai**, Yang Lu, Lei Xu, Linsey Pang, Richard Williamson, Hemanth Hegadehalii Madhavarao, Prakhar Mehrotra, Weidong Shi, "***CoJo-RAG: Conformalized Joint Co-Optimization of Retrieval and Fine-Tuning***", *AAAI 2027.*
43. **Rabimba Karanjai**, Suryabhan Singh Hada, Hemanth Hegadehalii Madhavarao, Libin N. George, Qun Gu, Wenhuan Sun, Neha Monga, Xiaojiao Yu, Richard Williamson, Prakhar Mehrotra, "***A Picture is Worth a Thousand Tokens: Lossless Multilingual Context Compression via Pictorial Intermediate Representations***", *AAAI 2027.*
44. Zhenghui Guo, Yilin Yang, Yuanbin Man, Miao Yin, Weidong Shi, **Rabimba Karanjai**, Omprakash Gnawali, Chengming Zhang, "***Allocation Before Ranking: Decoupled Token Compression for OmniLLMs***", c *NeurIPS 2026.*
45. Xinyu Hou, Yang Lu, **Rabimba Karanjai**, Pei-Chi Pan, Sen Lin, Lei Xu, Weidong Shi, "***Specialists Hold, Generalists Discount: Asymmetric Equilibrium in LLM Routing Auctions***", *NeurIPS 2026.*
46. **Rabimba Karanjai**, Yang Lu, Lei Xu, Weidong Shi, "***Closing the Cost Gap in Verifier-Guided RL: A Smart Contract Case Study***", *NeurIPS 2026.*
47. **Rabimba Karanjai**, Qun Gu, Hemanth Hegadehalii Madhavarao, Wenhuan Sun, Xiaojiao Yu, Suryabhan Singh Hada, Libin N. George, Uma Kona, Richard Williamson, Linsey Pang, Prakhar Mehrotra, "***CM-DPO: Constraint-Margin Direct Preference Optimization for LLM Planning***", *NeurIPS 2026.*
48. Yang Lu, **Rabimba Karanjai**, Dana Alsagheer, Lei Xu, Weidong Shi, "***Rules That Govern: A Per-Property Analysis of Multi-Agent LLM Deliberation***", *EMNLP 2026.*
49. **Rabimba Karanjai**, Yang Lu, Lei Xu, Weidong Shi, "***Profiler: Per-Question Persona Conditioning of Frozen LLMs via Retrieval and Fusion of Synthesized Profiles***", *EMNLP 2026.*
50. **Rabimba Karanjai**, Lei Xu, Yang Lu, Weidong Shi, "***VerifyGen-X: Secure Cross-Chain Smart Contract Generation via Scalable Reinforcement Learning from Formal Verification Feedback***", *NeurIPS 2026.*
51. Yilin Yang, Yuke Wang, **Rabimba Karanjai**, Weidong Shi, Chengming Zhang, "***Aligning Vision Language Models via anchor***", *NeurIPS 2026.*
52. **Rabimba Karanjai**, Yang Lu, Lei Xu, Weidong Shi, "***Unlocking On-Chain Intelligence: A Practical Framework for GenAI-Powered Smart Contracts***", *Middleware 2026.*
53. Dana Alsagheer, Yang Lu, Lei Xu, Weidong Shi, **Rabimba Karanjai** "***Bridging Confidentiality and Reliability: Open-Weight Agents for Legal Reasoning***", *ACM Symposium on Computer Science and Law 2026.*
54. **Rabimba Karanjai**, Yang Lu, Lei Xu, Weidong Shi, "***Hype or Hope? Training LLMs on Decentralized GPU Clouds***", *Middleware 2026.*

PATENTS

These patents were applied for when working in Industry.

- Mobile Multi-Party Digitally Signed Documents and Techniques for Using These Allowing Detection of Tamper - **US 16/372,593**
- Adaptive Heuristic Optimization Systems for Multi-Constraint Decision-Making Environments with Weighted Feedback Integration *Patent Pending*
- Asynchronous Event-Driven Transaction Orchestration and State-Tracking Architectures for Distributed Communication Networks *Patent Pending*
- High-Concurrency Serialized Processing Frameworks for Low-Latency Real-Time Signal Synthesis and Response Generation *Patent Pending*
- Distributed Resource-Aware Computational Framework for Privacy-Preserving Asynchronous Multi-Stream Inference *Patent Pending*
- Synchronous Context-Fusion Engine for Real-Time Multimodal Interaction via Spatiotemporal Environmental Perception *Patent Pending*

TALKS

INDUSTRY/CONFERENCE TALKS

1. “Beyond the Chatbot: A Blueprint for Trustable AI”, Google Deepmind, 2026
2. “Benchmarking vLLM on XLA Backends: From CUDA to OpenXLA for LLM Serving”, AI Sytems Devlabs, Google TPU Team, Google Mountain View, 2026
3. “Beyond the Chatbot: A Blueprint for Trustable AI”, Google Deepmind, 2026
4. “United in Defense: Architecting Safe and Trustworthy AI Agents”, BSides Seattle, 2026
5. “Unmasking the Shadows: AI Red Teaming in the Age of Gemini and VertexAI, fortified by SAIF”, KCDC 2025
6. “From Whiteboard to Users: Making Research Accessible”, Google I/O GDE Summit, 2025
7. “Unmasking the Shadows: AI Red Teaming in the Age of Gemini and VertexAI, fortified by SAIF”, Drexel University, 2025
8. “LLM, Reasoning and Agentic Gemma”, Google Korea, 2024
9. “Supercharging GenAI: Ray, Kubernetes, and TPUs for Lightning-Fast Inference”, KubeCon 2024.
10. “LLMinABox: On Device Personalized Diary & Concierge using your voice and Gemma”, Google Mt. View, 2024
11. “Mitigating Hallucinations in AI-Driven Medical Diagnosis”, Rice University, 2024.
12. “SolMover: Smart Contract Code Translation Based on Concepts”, Berkeley RDI, 2024.
13. “Privacy aware Zero Knowledge Login using Oauth2 and Passkey”, Columbia University, 2024.
14. “LLM Applications components and design patterns Hands on Workshop”, University Of Washington, 2024.
15. “Give your web apps superpower with Generative AI and Mediapipe”, University of Missouri - Kansas City, 2024.
16. “An Empirical Study of AI-based Smart Contract Creation”, Berkeley RDI, 2023.
17. “On Device Generative AI: Building your own Dall-E in the browser, welcome WebGPU”, Google San Jose, 2023.
18. “DeepSpeech: A Journey to ¡10% Word Error Rate”, Google Mt. View, 2023, NC Chapell Hill, 2023.
19. “Visualize your Data in a 3D VR world using A-Frame in WebVR”, OpenVis Conf, 2018
20. “Turning sensors into signals”, MIT, 2017
21. “Hardening Your IoT Endpoints: A Preventive Toolkit”, LinuxCon ContainerCon, 2017
22. “Optimizing Web Virtual Reality”, Web3d, 2017
23. “SecurityPI: IronClad your Raspberry PI ”, Linux Foundation Open IoT & ELC 2017, 2017
24. “State of WebVR & aframe: Yesterday, Today, and Tomorrow Beyond Horizon”,Open Networking Summit 2017.

GRANTS

These Grants were applied for by industry, as they are the only place a graduate student can apply as a primary investigator.

- **Sui Academic Research Award (2025)** - *Primary Investigator* - $25,000 - *Starts August,25*
- **Sui Academic Research Award (2024)** - *Primary Investigator* - $25,000
- **Sui Academic Research Award (2023)** - *Primary Investigator* - $25,000
- **Grant for Web** - *Primary Investigator* - $15,000

PROFESSIONAL ACTIVITIES

- NVIDIA Developer Champion, 2026
- Google TPU Builders, 2026
- Scientific Advisory Committee Member from University Of Houston, Texas Quantum Initiative, 2025
- Google Cloud Research Innovator, Google Research, 2024.
- Google Cloud Champions Innovator - AI & ML, 2023-2025.
- Google Developer Expert - AI, Google Cloud, 2024-2025 *and* Web, Chrome Team, 2018-2024
- Certificate of Excellence in Artificial Intelligence - University Of Houston
- Mozilla Tech Speaker, Emerging Technologies, 2019
- Mentor, Google Solutions Challenge, Google, 2024
- Mentor, Women Developer Academy, Google, 2024
- Judge, MIT Solve, 2020

SERVICE

- Associate Chair - *AAAI 2027, The 27th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)*, Track Chair – ICEDEG 2026
- Reviewer (Journals) - *PLOS One, Frontiers in Blockchain, ACM Distributed Ledger Technologies: Research and Practice, ACM Transactions on Asian and Low-Resource Language Information Processing, Connection Science, ACM Transactions on the Web*
- Reviewer (Conferences) - *CHI 2026, IEEE VR 2026, AAAI 2025, ICLR 2024, IEEE ICEDEG 2025*

LANGUAGES

Proficient in English, Bengali and Hindi.

REFERENCES

FROM ACADEMIA

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FROM INDUSTRY

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Dr. Lars Bergstrom
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